

In one line: Operators are ordinary (usually static) functions with special syntax; a **precedence group** decides how an unparenthesised expression *parses*. Subscripts are property-like members called with []. A literal has no type until context picks an ExpressibleBy...Literal type. ~= is what switch calls per pattern.

Stdlib precedence, high → low (a partial order)

BitwiseShift	<< >>	none
Multiplication	* / % &	left
Addition	+ - ^	left
RangeFormation	.. .< ...	none
Casting	is as as? as!	none
NilCoalescing	??	right
Comparison	== < == ~==	none
LogicalConjunction	&&	left
LogicalDisjunction		left
Ternary	? :	right
Assignment	= +=	right, assignment

23**2 — associativity decides**

right (math)

$2^{3^2} = 512$

left

$(2^3)^2 = 64$

none (the default): compile error

higherThan (orange box): above * and all below it; vs << unordered → a < b < a**c won't compile

Default (grey box): a group-less operator: above ?: only — a+++b+c = "unordered precedence groups"

C differs: & binds like *, | like +; a<bcc is an error (none)

switch v { case p: } runs p ~=: pattern left, value right, returns Bool

How it works

- **Declare, then implement**: infix operator `**`:
PowerPrecedence at file scope; implement as static func in a type (found by overload resolution via the operand types) or a global func. prefix/postfix funcs carry the modifier; infix ones don't.
- **Whitespace decides fixity**: space on both sides or neither = infix; `-x` prefix; `x!` postfix. `a -b` is *not* subtraction.
- **Reserved**: `=`, `>`, `..`, `?:`, `//`, `/*`, prefix `&`, postfix `? !`. `&&/||` are overloadable — their right side is an autoclosure, which is how they short-circuit.
- **precedencegroup**: `higherThan/lowerThan`, associativity (`left/right/none`, default `none`), `assignment`: `true` (folds into optional chains like `=`).
- **Subscripts**: parameters have **no argument labels** unless you name one (`subscribe(row r: Int) → g[row: 1]`); several params, variadic, defaults, **generic** (4.0), **static** (5.1, `Color("brand")`), `get-only` may be `async throws` (5.5). No `set-only` subscripts.
- **Literals**: the compiler calls the literal init of the *contextual* type, else the default (`Int`, `Double`, `String`, `Array...`).
- **callAsFunction** (5.2, SE-0253): makes `value(args)` work; overloads, labels, mutating, `async` allowed. `SwiftUI.dismiss()` is a `DismissAction` value. `@dynamicCallable` = the untyped, variadic cousin (scripting interop).
- `==`: `stdlib` has it for `Equatable` (`==`) and `RangeExpression` (contains); overload it to use a type as a case pattern.

Protocol (ExpressibleBy...)	required init
IntegerLiteral · FloatLiteral	init(integerLiteral:) · (floatLiteral:)
BooleanLiteral · StringLiteral	init(booleanLiteral:) · (stringLiteral:)
ArrayLiteral	init(arrayLiteral: Element...)
DictionaryLiteral	init(dictionaryLiteral: (K, V)...) · init(stringInterpolation:)
StringInterpolation	
NilLiteral — for Optional only	init(nilLiteral: ())

Example – operators + subscripts

```
precedencegroup PowerPrecedence {
    higherThan: MultiplicationPrecedence
    associativity: right }
infix operator ** : PowerPrecedence
func ** (b: Double, e: Double) -> Double { pow(b, e) }
struct Vec: Equatable { var x, y: Double
    static func + (l: Vec, r: Vec) -> Vec {
        Vec(x: l.x + r.x, y: l.y + r.y) }
    static func += (l: inout Vec, r: Vec) { l = l + r }
    static prefix func - (v: Vec) -> Vec { Vec(x: -v.x, y: -v.y) } }
struct Grid { var cells: [Int]; let w: Int
    subscript(r: Int, c: Int) -> Int { // g[1, 2]
        get { cells[r * w + c] }
        set { cells[r * w + c] = newValue } }
    subscript(row r: Int) -> ArraySlice<Int> { // g[row: 1]
        cells[r * w ..< (r + 1) * w] } }
```

Example — literals, callAsFunction, ~=

```
struct Money: ExpressibleByIntegerLiteral {
    let cents: Int
    init(integerLiteral v: Int) { cents = v * 100 } }
let price: Money = 5                                // 500 cents
struct Scale { let k: Double
    func callAsFunction(_ x: Double) -> Double { x * k } }
let twice = Scale(k: 2); twice(21)                 // 42.0
struct Prefix { let s: String }
func ~= (p: Prefix, v: String) -> Bool { v.hasPrefix(p.s) }
switch path {
case Prefix(s: "/api/"): routeAPI()                // Prefix ~= path
default: routeWeb() }
if case 200...300 = status { ok() }                 // Range ~= status
```

Traps

- ! A custom infix operator with no group gets DefaultPrecedence: mixing it with + or == unparenthesised does not compile.
- ! Forgetting associativity: right on a power operator: the default none rejects 2**3**2; left silently gives 64.
- ! A free == without : Equatable is invisible to generics, Set, firstIndex(of:); a free < gives no sort() — conform to Comparable.
- ! let s: Set = [1, 1, 2] silently has 2 elements; a dictionary literal with a duplicate key **traps at runtime**.
- ! Overloads + literals multiply the type checker's search: “unable to type-check this expression in reasonable time” — add type annotations, split the expression.
- ! Don't adopt ExpressibleByNilLiteral in your types — Apple reserves it for Optional.
- ! Operators nobody can read (<=>, |>) cost more than they save — only for a real algebra (vectors, matrices, parsers).

Remember

“Group parses, function computes; subscripts have no labels; literals wait for a type; case p means $p \approx v$.”